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Artificial Intelligence - Lab Session 1: Fundamentals

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1 Aim and evaluation

We compare different regression and classification methods on Concrete Compressive Strength, Superconductivity and CIFAR-10. Scaling and PCA are fitted using training data only. Validation data is used for model selection, and test data for the final evaluation. RMSE penalizes large regression errors more than MAE; classification accuracy measures the proportion of correct predictions, while macro F1 gives equal weight to each class.

2 Regression

2.1 Concrete Compressive Strength

The dataset contains 1,030 samples and eight input features. We use 721 samples for training, 206 for validation and 103 for testing. Training correlations with strength are strongest for *cement* ($r = 0.497$), *superplasticizer* ($r = 0.377$), *age* ($r = 0.328$) and *water* ($r = -0.301$). Zero superplasticizer values are retained: they can represent a mixture without this additive, rather than an outlier.

Polynomial degree and Ridge regularization are selected by cross-validation (n=5) on the training set. Table 1 compares the models on validation data.

Table 1: Concrete validation results.

Model	MAE (MPa)	RMSE (MPa)	R^2
Linear regression	8.47	10.65	0.60
Polynomial regression, degree 2	5.99	7.81	0.79
Polynomial Ridge, degree 3, $\alpha = 10$	4.95	6.51	0.85
Huber	8.32	10.90	0.58
RANSAC	10.32	18.65	−0.22

Polynomial Ridge achieves the lowest validation RMSE. Its test MAE is 5.20 MPa, RMSE is 7.01 MPa and $R^2 = 0.798$. Polynomial features capture nonlinear relationships, while Ridge limits large coefficients. Huber slightly improves MAE over linear regression but increases RMSE. RANSAC performs worse on this split; robustness to outliers alone does not capture nonlinear relationships.

2.2 Superconductivity

This dataset contains 21,263 samples and 81 features. The training, validation and test sets contain 14,884, 4,252 and 2,127 samples. Linear regression obtains the lowest validation RMSE (17.92 K), although tuned Ridge is practically tied (17.93 K). PCA retaining 95% of input variance reduces the data to 17 components, but PCA followed by Ridge gives an RMSE of 21.99 K. Tuning PCA and Ridge by cross-validation (n=3) improves this to 18.97 K, using 50 components and $\alpha = 10$, still above linear regression.

With a fixed validation set, increasing the training size from 500 to 14,884 reduces linear regression RMSE from 19.21 to 17.92 K. Gains become smaller after 2,000 samples. The selected linear model obtains a test RMSE of 17.44 K and $R^2 = 0.733$. Preserving input variance with PCA does not necessarily preserve the information most useful for predicting temperature.

3 CIFAR-10 classification

CIFAR-10 contains 60,000 RGB images of 32×32 pixels in ten classes. We select 10,000 training and 2,000 validation images from the official training set, keeping the official 10,000 test images separate. We compare k-NN ($k = 5$), logistic regression ($C = 1$) and SVM with an RBF kernel. SVM parameters C and γ are selected by cross-validation (n=3) on training subsets; the selected models are then fitted on all 10,000 training images.

For pixel features, scaling and PCA reduce the 3,072 inputs to 64 components. GIST descriptors are computed for all images using the supplied implementation: 16 Gabor filters and a 4×4 spatial grid give 256 features per grayscale image. Both representations use the same training and validation images. Table 2 reports test accuracy.

Table 2: CIFAR-10 test accuracy for the two representations.

Model	Pixels + PCA	GIST
k-NN	33.5%	49.7%
Logistic regression	37.8%	53.8%
SVM	46.5%	62.5%

GIST improves test accuracy by about 16 percentage points for each classifier. The GIST SVM, selected on validation data, obtains 62.5% accuracy and 0.625 macro F1. Its per-class F1 is higher for automobile (0.748) and truck (0.745) than for cat (0.422) and bird (0.494). The notebooks show examples of correct and incorrect predictions. GIST summarizes texture and spatial structure, but this grayscale representation loses colour and fine detail. These results compare the complete feature pipelines, including their different dimensions and preprocessing.

4 Conclusions

Regularized polynomial features improve concrete strength prediction. For superconductivity, more training data reduces error, while the tested PCA models do not improve on linear regression. For CIFAR-10, GIST gives higher accuracy than pixels reduced by PCA, and SVM performs best among the tested classifiers.